INCH-POUND
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SUPERSEEDING
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COMMERCIAL ITEM DESCRIPTION

JOINT DEPLOYABLE REFUELING PUMP

The General Services Administration has authorized this commercial item description for use by federal agencies.

1. SCOPE

1.1 Scope. This commercial item description (CID) describes the Joint Deployable Refueling Pump (JDRP) a trailer mounted fuel pumping system designed to issue fuel to an aircraft from an external fuel storage system. The unit is also capable of reverse operation.

2. SALIENT CHARACTERISTICS

2.1 JDRP description. The JDRP is a trailer mounted pumping system that can receive and issue fuel from a fixed source to perform aircraft refueling/defueling. The JDRP pumping system consists of a single pump driven by a multi-fuel engine controlled by an easily accessible operator panel. The JDRP is equipped with a storage rack for deployable accessories.

2.2 Design and construction. The design shall promote cost-effective, life-cycle sustainability by addressing considerations such as incorporating open standards, reducing pollutant emissions and wastes, and increasing fuel economy, while satisfying system performance requirements. It shall be designed and constructed so that no parts will work loose in service, and to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service. It shall be weatherproof and designed to prevent the intrusion of water, sand, and dust into critical operating components.

2.2.1 Materials, protective coatings, and finish.

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2.2.1.1 Protective coatings. Materials that deteriorate when exposed to sunlight, weather, or operational conditions normally encountered during the service life of the JDRP shall not be used or shall have means of protection against such deterioration that does not prevent compliance with the performance requirements specified herein. The JDRP shall be constructed of corrosion resistant materials to the maximum extent possible. Surface treatments, pre-treatments, and coatings shall be applied to all components that are not corrosion resistant. Surface treatments include electroplated coatings, thermal sprayed coatings, and hard anodized coatings. Surface pre-treatments include thin film anodize, conversion coatings, zinc phosphate, and other coatings used to enhance subsequent corrosion protective coating adhesion and performance. Corrosion protective coatings include corrosion inhibited epoxy primers and polyurethane topcoats. Corrosion protective coatings that chip, crack, or scale with age or extremes of climatic conditions or when exposed to heat shall not be used. Exposed surfaces of fasteners, handles, and fittings shall also be primed and painted. Commercial items (see 6.3.4) used as components of the JDRP (but not the JDRP itself) may be prepared and coated in accordance with the manufacturer's standard practice, provided it is compatible with the exterior finish color. Some commercial items (see 6.3.4) such as axles, springs, wheels, backing plates/brake dust shields, towbars, etc. shall have the protective coatings required herein in accordance with the respective material substrates. When in contact with ferrous materials, steel fasteners shall be plated with zinc/nickel in accordance with MIL-PRF-32647. When in contact with aluminum materials, steel fasteners shall be coated with a Zinc/Aluminum Corrosion Protective Coating in accordance with ASTM F1136 Grade 3.

2.2.1.2 Surface preparation. Components shall be cleaned, degreased, and prepared prior to any scuffing, sanding, blasting, or application of any pretreatment. Surface preparation and pretreatment shall be in accordance with the respective primer and topcoat specifications. All corrosion protection steps, regardless of material, shall be applied before any new oxidation or rusting occurs post-treatment and shall be completed prior to application of any coating used as either primer or topcoat. Surfaces shall be scuffed, sanded, and/or blasted, depending upon material type and thickness, to remove any existing surface impurities such as primer/paint/topcoat, corrosion, welding/grinding/cutting slag, etc. that may prevent adhesion and protection of further corrosion control measure. Any surface undergoing a media blasting process shall have the blasting pressure, blasting material type, and blasting material grit suitably chosen to prevent warpage or other structural damage. Coatings requiring a “near-white” ferrous metal finish shall have the surface(s) prepared in accordance with SSPC-SP 10/NACE No. 2. General surface finish and cleanliness shall be in accordance with SSPC-Vis 1.

2.2.1.3 Aluminum pretreatment. Aluminum surfaces used in construction of the JDRP shall be treated with a suitable conversion coating prior to application of any other corrosion prevention coating. Anodizing or a MIL-DTL-81706 conversion coating process may be utilized, at the discretion of the JDRP manufacturer. Any anodizing shall be applied in accordance with MIL-PRF-8625, Type II, Class 1; no further conversion coating is required on anodized aluminum surfaces. Alternatively, aluminum surfaces shall have MIL-DTL-81706, Type II, Class 1A, Form I, Method C chemical conversion coating applied in accordance with MIL-DTL-5541, Type II, Class 1A.

2.2.1.4 Ferrous pretreatment. Ferrous surfaces used in construction of the JDRP shall be treated with a metal etching compound prior to application of any further pretreatment process(es).

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Ferrous surfaces with iron or zinc phosphate coatings/etches do not require further pretreatment. Ferrous surfaces without iron or zinc phosphate coatings/etches shall be treated with a wash primer in accordance with TT-C-490 to seal out flash rust.

2.2.1.5 Primer. Raw metal edges, to include fastener and drain holes, shall be coated with primer prior to applying the topcoat. Any surfaces where primer and/or topcoat is not required, or intended to be applied, shall be masked off using materials and processes suitable for use with applied primer and/or topcoat system. A primer color shall be chosen that is compatible with the color of the topcoat used such that scratches in the topcoat do not yield an obtrusive contrast with any exposed primer (tone-down primer in a similar shade/color family as the topcoat).

2.2.1.6 Ferrous surfaces. Ferrous structures and surfaces shall be primed with a non-water reducible zinc rich primer in accordance with MIL-DTL-53022, Type IV, Class L. The primer system shall be allowed to dry and fully cure in accordance with the primer manufacturer's directions prior to applying the topcoat.

2.2.1.7 Aluminum and mixed aluminum and ferrous surfaces. Aluminum and mixed aluminum and ferrous structures and surfaces shall be primed with an epoxy primer in accordance with MIL-DTL-53022, Type IV, Class L.

2.2.1.8 Topcoat. All JDRP components, except for commercial items (see 6.3.4), plastic, rubber, slide able (brake linkages), and friction surfaces (brake surfaces) shall be top coated with polyurethane in accordance with MIL-PRF-85285, Type IV, Class H. Chemical Agent Resistant Coating (CARC) shall not be used. The coating shall be free from runs, sags, orange peel, or other defects. The contractor shall utilize the topcoat material manufacturer's specifications to accomplish application of the coating system in accordance with the manufacturer's directions to the minimum dry film thickness required herein. As an alternative to the above topcoat and primer, a contractor-developed corrosion control approach, subject to government approval, may also be submitted for review.

2.2.1.9 Finish color. The exterior finish color of the JDRP and the inner surfaces of compartments, but not components installed in the compartments, shall be matte grey, Color Number 26173 of SAE AMS-STD-595.

2.2.1.10 Exclusion of water. The design of the JDRP shall be such as to prevent water leaking into, or being driven into, any part of the JDRP interior when either in an operating or travelling configuration. All panels, covers, etc., shall be provided with sealing arrangements such that the entry of water is minimized when these items are correctly closed. Particular care shall be taken to prevent wetting of equipment and heat and sound proofing materials. Sharp corners and recesses shall be avoided so that moisture and solid matter cannot accumulate to initiate localized attack. Sealed floors with suitable drainage shall be provided for storage compartments, engine compartments, and other areas in the JDRP that could collect and retain water.

2.2.1.10.1 Fluid traps and faying surfaces. There shall be no fluid traps on the JDRP. Faying surfaces of all structural joints, except welded joints, shall be sealed to preclude fluid intrusion.

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2.2.1.10.2 Ventilation. Ventilation shall be sufficient to prevent moisture retention and buildup.

2.2.1.10.3 Drainage. Drain holes shall be provided to prevent collection or entrapment of water or other unwanted fluid in areas where exclusion is impractical. All designs shall include considerations for the prevention of water or fluid entrapment and ensure that drain holes are located to effect maximum drainage of accumulated fluids. The number and location of drain holes shall be sufficient to permit drainage of all fluids when the JDRP is on a 10-degree longitudinal slope facing both up and down and on a 10-degree side slope in each direction (right and left side facing up the slope). The minimum size of the drain holes shall be 0.375 inch.

2.2.1.11 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall not be in contact with each other. Metal plating or metal spraying of dissimilar base metals to provide electromotively compatible abutting surfaces is acceptable. The use of dissimilar metals when separated by suitable insulating material is permitted, except in systems where bridging of insulation materials by an electrically conductive fluid can occur. Sealants or gel type gasket materials shall be used between faying surfaces and butt joints.

2.2.2 Markings. All external devices which require an operational or maintenance interface shall be marked in accordance with MIL-STD-130. Markings shall be applied with decals and shall be 1-inch-high block letters unless prohibited by the available space. In such cases, the markings shall be the largest size possible. Markings, Information/Caution shall be Black, Color Number 37038 of SAE AMS-STD-595, and Markings, Warning/Danger shall be Red, Color Number 31136 of SAE AMS-STD-595. The center of gravity of the JDRP shall be marked on the unit within 1-inch of the calculated center of gravity. Tire inflation pressure shall be marked as near to each tire as practicable.

2.2.3 Identification and information plates.

2.2.3.1 Identification plate. An identification plate in accordance with MIL-STD-130 shall be securely attached to the JDRP in a readily accessible location. The identification plate shall contain the following information: nomenclature, part number, serial number, date of manufacture, manufacturer's name, Commercial and Government Entity (CAGE) code, contract number, date of warranty expiration, National Stock Number (NSN). The JDRP and any of its components for which the Government's unit cost is more than $5,000, is serially managed, or the procuring agency determines is mission essential, shall have Unique Identification (UID) (also known as Item Unique Identification (IUID)) information permanently affixed on or near the respective identification plate(s), marked in accordance with MIL-STD-130. UID information shall be included as both a 2D Data Matrix and human readable markings.

2.2.3.2 Transportation data plate. A transportation data plate shall be securely attached to the JDRP in a readily accessible location. The plate shall contain at least the following information:

a. Side and rear silhouette views of the JDRP.

b. Horizontal and vertical location of the center of gravity of the JDRP in air transportable configuration, marked on the silhouette views.

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c. Shipping weight.

d. Loading cubage.

e. Overall height, width, and length.

f. Front and rear axle loads.

g. Tie down information.

2.2.3.3 Lubrication data plate. The JDRP shall be provided with a permanently-marked lubrication data plate constructed of a non-corroding metal, mounted on the exterior surface of the JDRP that shall direct attention to all lubrication fittings and components that require lubrication. The plate shall identify the type and grade of lubricant required for operational temperatures specified in 2.3.1.

2.2.4 Environment, Safety, and Occupational Health (ESOH).

2.2.4.1 Hazardous material. The design shall minimize and control hazards associated with the inclusion or use of hazardous or toxic materials and the generation of toxic or noxious gases. The JDRP shall not generate or use Class I or Class II Ozone Depleting Substances (ODS) during operation, maintenance, or disposal. Class I ODS and hazardous materials shall not be used in any system, component, or process. The JDRP shall not contain or use either hexavalent chromium or cadmium without written approval by the procuring activity. Hazardous materials are defined in DAFMAN 32-7002; Class I and Class II ODS are defined in 40CFR82.

2.2.4.2 Component protection. All space in which work is performed during operation, service, and maintenance shall be free of hazardous protrusions, sharp edges, or other features which may cause injury to personnel. All rotating and reciprocating parts and all parts subject to high operational temperatures or subject to being electrically energized, that are of such nature or so located as to be hazardous to personnel, shall be guarded or insulated to eliminate the hazard. All wires, cables, tubes, and hoses shall be supported and protected to minimize chafing and abrasion and shall be located so as to provide adequate clearance from moving parts and high operational temperatures. Grommets shall be provided wherever wires, cables, tubes, or hoses pass through bulkheads, partitions, or structural members.

2.2.4.3 Foreign object damage (FOD). All loose metal parts, such as pins or connector covers, shall be securely attached to the JDRP with wire ropes or chains. "Dog tag" style beaded chains shall not be provided. Removable panels, if provided, shall be attached with captive fasteners. Tire valve stem caps shall be made of plastic.

2.2.4.4 Fire extinguisher. A dry chemical fire extinguisher, Type 1, Class 1, Size 20 in accordance with A-A-393, shall be mounted on the exterior of the unit enclosure in a location easily accessible from the control panel.

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2.2.4.5 Noise. The design shall ensure that noise created by the JDRP minimizes exposure of personnel to noise hazards during operations and maintenance activities. The maximum average sound level shall not exceed limits defined in 3.11.1 of SAE ARP1274. If measured noise at any distance exceeds maximum average of 85 dBA, signs in accordance with 4.2 of MIL-STD-1474 shall be mounted on all four sides of the unit. All sound level measurements shall be made in accordance with 3.11.2 of SAE ARP1247.

2.2.4.6 Grounding Reel. The JDRP shall have a grounding reel that is spring retractable and 75 feet in length. The reel shall have both clamp and plug style grounding attachments.

2.2.5 Electromagnetic interference (EMI). The JDRP shall be in accordance with the following conducted and radiated emission and susceptibility requirements of MIL-STD-461: RE102, and RS103.

2.2.6 Human systems integration. The JDRP shall be designed in accordance with MIL-STD-1472 for ease of operation, inspection, and maintenance, including the use of arctic mittens and Mission-Oriented Protective Posture (MOPP) Level 4 Chemical Warfare Gear. Chemical Warfare Gear is not required for preventive maintenance or major corrective maintenance.

2.2.7 Fastening devices. All screws, bolts, nuts, pins, and other fastening devices shall be properly designed, manufactured, and installed with adequate means of preventing loss of torque or adjustment. Cotter pins, lock washers, or nylon patches shall not be used for this purpose, except for the attachment of trim items or as provided in commercial (see 6.3.4) components. Tapped threads shall have a minimum thread engagement in accordance with Table I.

TABLE I. Minimum thread engagement.

Material	Minimum Thread Engagement
Steel	1.0 times the nominal fastener diameter
Cast iron, brass, or bronze	1.5 times the nominal fastener diameter
Aluminum, zinc, or plastic	2.0 times the nominal fastener diameter

2.2.8 Welders and welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the welded parts. Welds shall transmit stress without cracking or permanent distortion when the parts connected by the welds are subjected to test, proof, and service loadings.

2.3 Environmental conditions.

2.3.1 Operating temperature range. The JDRP shall be capable of operating in ambient temperatures ranging from -40 to 140 °F.

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2.3.2 Storage temperature range. The JDRP shall be capable of being stored in ambient temperatures ranging from -65 to 160 °F.

2.3.3 Precipitation.

2.3.3.1 Rain. The JDRP shall be capable of storage and operation during rainfall of 4-inches per hour for two consecutive hours with winds of up to 35 knots impinging on the JDRP at angles from vertical to 45°.

2.3.3.2 Snow. The JDRP shall be capable of storage and operation during accretion of wet snow up to 2-inches per hour for at least 12 hours.

2.3.3.3 Ice. The JDRP shall be capable of storage and operation with ice accretion up to 1.5-inches on exposed horizontal surfaces. An operator may use an ice scraper for five minutes during the start-up process.

2.3.4 Solar radiation. The JDRP shall not be adversely affected by full time exposure to solar radiation, such as those conditions encountered in desert environments.

2.3.5 Fungus. All materials used in the JDRP shall be fungus resistant or shall be suitably treated to resist fungus. Materials treated for fungus resistance shall retain their original electronic and physical properties, shall not present toxic hazards, and treatment shall last for the entire service life of the part. The JDRP shall be suitable for operation and storage in conditions encountered in a tropical environment.

2.3.6 Salt fog. The JDRP shall be capable of storage and operation in high temperature, high humidity, salt laden, seacoast environments without damage or deterioration of performance.

2.3.7 Sand and dust. The JDRP shall be capable of storage and operation during exposure to wind-blown sand or dust without damage or deterioration of performance.

2.4 Weight and dimensions. Overall weight and dimensions in air transport configuration (see 2.5.2) shall not exceed two 463L pallet positions

Weight	7500 pounds (dry weight), Weight of JDRP fully loaded with fuel and 1,500 pounds of accessories must be under 10,000 pounds.
Length	208 inches.
Width	84 inches.
Height	96 inches.

2.5 Transportability.

2.5.1 Surface transportability. The JDRP shall be transportable via all modes of surface shipment (highway, rail, and water) in accordance with MIL-STD-1366, and shall be capable of withstanding the mechanical shock and vibration characteristics of highway, rail, and water transport, except that design for rail impact testing (see 5.2.5 of MIL-STD-1366) is not required.

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2.5.2 Air transportability. The JDRP fully loaded with 1,500 pounds of accessories (see 2.11) shall be transportable on C-5, C-17, and C-130 aircraft in accordance with MIL-STD-1791. The JDRP shall fit on two 463L aircraft pallets. In all air transport configurations, the JDRP shall be capable of being restrained and withstanding, without loss of serviceability, 2.0 G up and 4.5 G down accelerations, and shall be capable of being restrained and withstanding, without loss of structural integrity, 3.0 G forward, 1.5 G aft, and 1.5 G lateral accelerations. The JDRP shall be equipped with pressure relief devices or configured for air transport to prevent any part from becoming a projectile in the event of catastrophic loss of aircraft cabin pressure. The JDRP shall roll on and off the aircraft, negotiating the required maximum ramp angles without shoring.

2.5.2.1 Shoring. The JDRP shall be air transportable without shoring.

2.5.2.2 Tire pressure. Tire pressure shall not exceed 100 pounds per square inch (psi) and shall not be reduced for air transport.

2.5.2.3 Winch points. Winch points shall be provided at the front and rear of the JDRP for aircraft loading using the aircraft's winch.

2.5.2.4 Equipment removal and reconfiguration. Preparation for air transport shall take no more than 1 man-hours and restoration to operating configuration shall take no more than 1 man-hours for 2 persons using common hand tools (see 6.3.5). All equipment removed shall be stored on the JDRP; caps and plugs shall permit moving and storage in transport configuration.

2.5.3 Tie downs. The JDRP shall be symmetrically restrained during air and ground transport. Tie down points shall be marked for capacity, with a clear opening compatible with MIL-DTL-25959 and MIL-PRF-27260 tie down devices. Each end of each tie down device shall terminate at a tie down point and not pass through any other tie down point. There shall be no interference between tie down devices and the JDRP. The tie down provisions shall be designed in accordance with MIL-STD-209.

2.5.4 Lifting provisions. The JDRP shall be equipped with sufficient attachment points so located that it can be lifted by crane; each attachment point shall be marked "Lift Point". The lifting provisions shall be in accordance with 5.1 through 5.1.4 of MIL-STD-209.

2.6 Maintainability. The JDRP shall be designed for maintainability in accordance with 5.9 through 5.9.18 of MIL-STD-1472; forces shall not exceed those specified for both males and females.

2.6.1 Preventive maintenance. The recommended preventive maintenance interval (PMI) shall be at least 200 operating hours or more if achievable. Preventive maintenance tasks shall not require more than 4 man-hours.

2.6.2 Inspection and servicing provisions.

a. Pre-use inspections and servicing tasks shall not require tools.

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b. Routine service and preventive maintenance shall not require special tools (see 6.3.10).

c. Drain plugs and filters shall be directly accessible from the ground and oriented to have unimpeded drainage to a catch pan.

d. The JDRP shall be designed with maximum usage of sealed lifetime lubrication bearings.

2.6.3 Special tools. The design of the item shall minimize the requirement for special tools (see 6.3.10). All special tools shall be provided with, and stored on, the JDRP.

2.6.4 Diagnostic software. A copy of any diagnostic software required or recommended for maintaining the JDRP shall be provided with each JDRP on CD-ROM or DVD-ROM.

2.6.5 Service life. The JDRP shall be designed for a minimum service life of 15 years without a need for major overhaul or repair. Service life does not apply to wear items such as tires, brakes, lights, hoses, cables, etc. Accomplishment of manufacturer recommended and defined service tasks and intervals shall maintain output performance in excess of 90% of design through the entire service life. Additionally, the JDRP shall be suitable for extended storage in an outside environment for 15 years.

2.7 Performance. The JDRP shall be capable of receiving and issuing fuel from a fixed source to perform aircraft refueling at a minimum rate of 300 Gallons Per Minute from a single-issue point. The JDRP shall be capable of drawing fuel from a minimum of 100 feet away and issue fuel a minimum of 75 feet away. The JDRP single-point nozzle pressure shall not exceed 55 psi during operation. The JDRP shall be compatible with pumping aviation fuel (JP-8, F-24, JA, JA1, JP-5, and TS1 Fuels), Automotive Diesel, and Arctic Fuel grades.

2.8 Mobility. The JDRP shall be in accordance with SAE AS8090 for Type III, Group A mobility except as otherwise specified herein. The JDRP must be DOT compliant for on road operation.

2.8.1 Terrain.

2.8.1.1 Paved surfaces. The JDRP shall be towable at speeds up to 60-miles per hour (mph) over paved surfaces.

2.8.1.2 Graded gravel. The JDRP shall be towable at speeds up to 25-mph over graded gravel.

2.8.2 Towing force. The towing force shall be in accordance with 3.3.4 of SAE AS8090.

2.8.3 Clearances. Ground clearance, ramp breakover angle, and angles of approach and departure shall be in accordance with 3.3.9 of SAE AS8090.

2.8.4 Suspension. The JDRP shall have a suspension system in accordance with 3.4.1.2 of SAE AS8090.

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2.8.5 Brakes.

2.8.5.1 Service brakes. The JDRP shall have service brakes in accordance with 3.6.2 of SAE AS8090.

2.8.5.2 Leveling Jacks. The JDRP shall have leveling jacks on all 4 corners at minimum. Leveling jacks shall prevent any movement while in use.

2.8.6 Bumper. The JDRP shall have a rear bumper in accordance with 3.7 of SAE AS8090.

2.8.7 Fenders. The JDRP shall have fenders in accordance with 3.9 of SAE AS8090.

2.8.8 Mudflaps. The JDRP shall have mudflaps in accordance with 3.12 of SAE AS8090.

2.8.9 Pintle hook. The rear pintle hook, specified in 3.13 of SAE AS8090, shall not be provided; the JDRP shall be towed singly, not in trains, as specified in 3.3.8 of SAE AS8090.

2.8.10 Towbar. The JDRP shall be equipped with a towbar in accordance with 3.17.1 of SAE AS8090. It shall have a lunette eye in accordance with A-A-52464 and accommodate any pintle hook height from 22 through 30 inches.

2.8.11 Tires and wheels. The JDRP shall be equipped with single or dual tubeless steel belted radial tires with on/off-road type tread mounted on steel disc wheel assemblies. Tire and wheel assemblies shall be identical at all wheel positions. Tires and wheels shall be in accordance with the *Tire and Rim Association Year Book* requirements for this application.

2.9 Engine and related equipment. The JDRP shall be equipped with a multi-fuel engine. The engine shall be certified to comply with the most stringent Environmental Protection Agency (EPA) non-road engine emission requirements possible while meeting the requirement to operate on fuels containing over 15 parts per million (ppm) sulfur.

2.9.1 Engine.

2.9.1.1 Engine specifications. The JDRP shall contain an air cooled, multi-fuel internal combustion engine capable of meeting operating requirements in 2.7. The engine shall be capable of operating without lugging, stalling, or over speeding during operation in any of the environments mentioned in 2.3.

2.9.1.2 Engine throttle control. The JDRP shall utilize a manual throttle to control engine RPM. The throttle control shall allow for easy RPM increase and decrease, lockable at a desired setting, and capable of rapid shutdown with ease. Throttle control shall be the main component to manage fuel flow.

2.9.1.3 Start Switch. The JDRP shall provide a start switch that serves as the primary means to energize components for operation and serves as the engine starting interface. The start switch

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shall electronically energize all components required for operation and provide an engine start and run position. The start switch shall be of a hermetically sealed design.

2.9.1.4 Starter. The engine shall be equipped with a 12-volt direct current (DC) or 24-volt direct current (DC) electric starter.

2.9.1.5 Engine starting aids. The engine shall start within 15 seconds cranking in any ambient temperature within the required operating range of the JDRP. Internal engine starting aids, and fluid starting aids may be used prior to and during the start period to facilitate engine starting under the following conditions:

TABLE II. Engine starting aids.

Temperature Range	Starting Aids Permitted
40 °F through 140 °F	None
0 °F through 39 °F	Internal engine starting aids and fluid starting aids
-45 °F through -1 °F	Internal engine starting aids and fluid starting aids.

2.9.2 Engine air intake system. The engine air intake system shall be in accordance with 3.13.1.4.3 of SAE ARP1247. The inlet shall not draw air from directly beneath the JDRP and shall not be located near the cooling system air outlet nor the engine exhaust outlet. Joints shall be minimized between the air filter outlet and the actual engine air inlet and shall be designed to ensure no leakage of unfiltered air into the engine. A differential pressure air filter service indicator shall be provided.

2.9.3 Engine lubrication system. The engine lubrication system shall be designed so that the JDRP can be operated on a 10-degree longitudinal slope facing both up and down and on a 10-degree side slope in each direction (right and left side facing up the slope).

2.9.4.1 Engine oil. The engine shall be compatible with Grade 15W40 of MIL-PRF-2104 from 0 to 140 °F. Oil pre-heat for operation below 0 °F is allowed. The engine shall be compatible with arctic engine oil in accordance with MIL-PRF-46167 from -45 to 60 °F.

2.9.4.1.1 Engine oil operating temperature. The engine oil sump temperature shall not exceed 250 °F or the engine manufacturer's recommendations at 140 °F.

2.9.4.1.2 Engine oil consumption. The engine oil consumption shall not exceed 0.0035 pounds per brake horsepower-hour (lbs/bhp-hr) under any operating condition.

2.9.4.2 Engine oil filter. The engine oil filter shall be in accordance with 3.13.1.4.4 of SAE ARP1247.

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2.9.5 Exhaust system. The exhaust system shall be constructed of stainless steel. The muffler(s) shall be constructed of aluminized steel or stainless steel. Exhaust system outlet(s) shall be directed away from personnel accessing any control panel or equipment compartment and the engine air intake, and shall not be directed toward the ground.

2.9.6 Engine fuels and fuel system.

2.9.6.1 Engine primary fuels. The following shall be primary fuels (see 6.3.9):

a. JP-5, in accordance with MIL-DTL-5624, -45 to 140 °F ambient air temperature.

b. JP-8, in accordance with MIL-DTL-83133, -45 to 140 °F ambient air temperature.

c. F-24, in accordance with NATO AFLP-3747, -45 to 140 °F ambient air temperature.

2.9.6.2 Engine alternate fuels. The following shall be alternate fuels (see 6.3.1):

a. 1-D S15, 1-D S500, 1-D S5000 in accordance with ASTM D975, below 60 °F ambient air temperature.

b. 2-D S15, 2-D S500, 2-D S5000 in accordance with ASTM D975, above 32 °F ambient air temperature.

c. TS-1 in accordance with GOST 10227-86 with JP-8 additives, -45 to 140 °F ambient air temperature.

2.9.6.3 Engine emergency fuels. The following shall be emergency fuels (see 6.3.7): Jet A, Jet A1 in accordance with ASTM D1655-05, all ambient temperatures.

Note: Jet A and Jet A1 may contain corrosion inhibitor/lubricity improver and icing inhibitors which shall result in properties equal to JP-8.

2.9.6.4 Fuel system. The fuel system shall be in accordance with 3.13.1.5.1 through 3.13.1.5.11 of SAE ARP1247 except as otherwise specified herein. The fuel system shall be constructed of materials which are compatible with the fuels listed in 2.9.6.1 and 2.9.6.2. Copper shall not be used in the fuel system. The fuel system shall be equipped with a fuel shut-off valve(s) to prevent continuous spillage when fuel lines are disconnected for service.

2.9.6.4.1 Fuel priming pump. The JDRP shall be equipped with an electric fuel pump in addition to the mechanical fuel pump. The electric pump shall be used as a priming pump capable of re-priming the engine fuel system following fuel exhaustion.

2.9.6.4.2 Fuel filters. Primary and secondary fuel filters and a heated fuel/water separator shall be provided. The fuel/water separator shall include a water coalescer and a drain valve that is readily accessible by an operator or a mechanic. A combination fuel filter and fuel/water separator may

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be provided. Fuel filter elements shall be easily replaceable by a mechanic using nothing more than common hand tools (see 6.3.5) without loss of engine prime.

2.9.6.4.3 Fuel tank. The fuel tank shall be in accordance with 3.13.1.5.5 through 3.13.1.5.9 of SAE ARP1247. The tank shall be designed so that the JDRP can be operated on a 15-degree longitudinal slope facing up and down and on a 15-degree side slope in each direction (right and left side facing up the slope). The tank shall be provided with corrosion protection. A 0.25- to 0.375-inch nominal drain valve shall be provided for emptying fuel and sediment into a container underneath the JDRP without removal of the tank or any other major component. The fuel tank shall have a fuel fill opening of not less than three inch inside diameter and shall be designed to drain fuel spillage overboard for collection outside the JDRP. The fuel cap shall be equipped with a retention device to prevent loss and FOD. The fuel fill opening, fuel cap, and fuel cap retention device shall be fabricated from non-sparking material. The fuel tank shall have a minimum 5-gallon capacity. The fuel tank shall contain a float valve to regulate fuel feed line to a maximum of 75% tank fill level.

2.9.6.4.4 Fuel feed line. The JDRP shall feature a fuel feed line to redirected pumped fuel as needed to maintain tank fill level. A fuel feed line shall be installed between fuel pump discharge manifold and the fuel tank and be equipped with a manual shutoff ball valve. The fuel feed line shall be regulated to limit tank fill to 75% capacity during issue/receive operations.

2.9.7 Engine diagnostic and emergency shutdown systems.

2.9.7.1 Engine emergency shutdown system. The engine shall be equipped with an engine emergency shutdown system consisting of a fuel cutoff solenoid activated by the following conditions:

a. Low oil pressure, less than 10 psi or in accordance with the engine manufacturer's recommendations.

b. Overheat condition in accordance with the engine manufacturer's recommendations.

c. Engine over speed should the engine exceed 110% rated speed. This is required for engines over 100 bhp and optional for engines less than 100 bhp rated horsepower.

2.9.8 Operator control panel. The JDRP operator control panel shall be readable from 15 feet during daytime or nighttime operations. The panel shall include a light with a minimum of 250 lumens and a maximum of 450 lumens. The light shall have a dimmer switch capable of completely blacking out the control panel. The operator control panel shall be in accordance with 3.13.1.8 of SAE ARP1247. The operator control panel shall provide the operator with the following instruments and controls:

a. Instruments.

i. Tachometer.

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ii. Engine temperature gauge.

iii. Oil pressure gauge.

iv. Fuel level gauge.

v. Hour meter.

vi. Pump output pressure.

vii. Nozzle Pressure/Single point outlet.

viii. Flow rate.

ix. Battery volt meter.

b. Controls

i. Manual throttle control.

ii. Start switch.

iii. Master battery disconnect.

iv. Emergency shut-off switch.

v. Variable dimmer switch for panel lighting.

2.9.9 Deadman control. A deadman control shall be provided with the JDRP that shall start and stop flow from the hose outlet. The deadman function shall be a squeeze handle type device. Deadman control shall be 75 feet in length and on a retractable reel.

2.10 Electrical system. The JDRP shall have a 12-volt or 24-volt, negative ground electrical system in accordance with 3.13.1.2 of SAE ARP1247 except as otherwise specified herein.

2.10.1 Alternator. A single or dual alternator charging system in accordance with 3.13.1.4.9 of SAE ARP1247 shall be provided. The alternator shall be capable of restoring the energy expended during an engine start in less than 15 minutes of engine idle at -40 °F.

2.10.2 Battery and battery compartment.

2.10.2.1 Batteries. Batteries shall be of the commercial (see 6.3.4) maintenance-free sealed lead acid, starved electrolyte, gas recombination, spiral wrapped, absorbent gas mat (AGM), top post type in accordance with MIL-B-18013/1.

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2.10.2.2 Battery compartment. The batteries shall be enclosed in a corrosion-resistant, weatherproof box or compartment and shall be readily accessible.

2.10.2.3 Battery cables. The battery cables shall be sized to handle the system voltage and current levels, be clearly identified with "+" and "-" or red and black markings, and shall not be spliced.

2.10.2.4 Master battery disconnect. A master battery disconnect switch shall be installed to allow complete isolation of battery from any electrical loads.

2.10.3 Plugs and connectors. The JDRP shall be provided with an electrical plug in accordance with ISO 1724 and cable of sufficient length for tow vehicle connection. The plug shall be located near the trailer tongue.

2.11 Equipment/hose storage rack. The JDRP shall include a top mounted equipment storage rack rated for 1,500 pounds. The rack shall allow for organized storage of all hoses and equipment provided with JDRP. The rack shall be provided with adequate tie down points to secure items during transport.

2.12 Storage. The JDRP shall provide sufficient storage space for all tools, adaptors, spare parts, and repair kit items.

2.13 Fuel pumping system.

2.13.1 Pumping system. The fuel-servicing unit shall be comprised of a pump that is self-priming, and engine driven. When adequate Net Positive Suction Head (NPSH) is available, the pump shall have the ability to deliver fuel to an aircraft through one single point nozzle at min 300 GPM. The JDRP shall be capable of drawing fuel from a minimum of 100 feet away and issue fuel a minimum of 75 feet away. The JDRP shall have an analog flow meter with a 5-digit resettable totalizer and an 8-digit non-resettable totalizer that is compatible with various fuels measured in gallons. Flow meter must be accurate to ± 1%.

2.13.2 System valves. All fuel valves shall be quarter turn wafer-style butterfly valves.

2.13.3 Piping manifold concept. All operations shall be configured using valve positioning to direct fuel flow. System piping shall provide low point drains, culminated at a single outlet point, allowing for maximum fuel draining for reconstitution, purge, and/or storage.

2.13.4 Hydrostatic system pressure. The piping system and all system components shall be designed to withstand 1.5 times maximum pump pressure. The system shall also withstand any pressure spike/surges with automatic pressure relief and include a manual relief system should the automatic reliefs fail.

2.13.5 Fuel filtration. The unit shall be equipped with a minimum of 300 gallons per minute filter separator using the most current Energy Institute standards. Filtration systems used shall meet or exceed EI 1581 7th edition, category M, type S requirements. The filter separator must be able to be configured using piping and valves so fuel can be filtered during receipt and issue operations.

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Filter shall be easily accessible to allow for servicing of filter elements and coalescer cartridges. The filter separator shall be equipped with a differential pressure gauge.

2.13.6 Water slug control. A water slug control system shall be integral to the filter separator. The slug control will shut off fuel flow until water is manually drained from filter separator.

2.13.7 Fuel sampling. Fuel sampling ports shall be incorporated upstream and downstream of filter separator. Fuel sampling ports shall extend half the diameter of into the pipe to allow fuel samples to be taken at the center of the pipe. The fuel sampling ports shall be equipped with quarter-turn ball valves to allow for fuel sampling. The fuel sampling ports shall allow adequate space for connection of a Millipore Filter Corporation, Gammon Technologies or equivalent fluid sampling device.

2.13.8 Pumping Controls. The JDRP shall have operator-selected pumping modes as follows.

a. Fuel Servicing - High Pressure mode. The JDRP shall (1) control the flow rate by varying the engine RPM, and (2) adjust the flow rate to prevent single point nozzle pressure from exceeding 50 ± 5 psi. In the event the secondary pressure control is activated due to the failure of the primary pressure control system.

b. Fuel Servicing - Low Pressure mode. The JDRP shall (1) control the flow rate by varying the engine RPM, and (2) adjust the flow rate to prevent the single point nozzle pressure from exceeding 30 ± 5 psi. In the event the secondary pressure control is activated due to the failure of the primary pressure control system.

c. Defuel mode. The JDRP shall defuel fuel from a host aircraft at a reduced rate with the assistance of aircraft boost pumps.

2.13.9 Nozzle pressure. When using a fuel-servicing mode to deliver fuel through one or both discharge hoses, the primary nozzle pressure control system shall limit the nozzle pressure to 30 ± 5 psi when in low pressure mode and 50 ± 5 psi when in high pressure mode. A secondary pressure control system for each discharge hose shall limit the nozzle pressure to 55 psi when in either pressure mode. Hose end pressure control valves are allowed for secondary pressure control.

2.13.10 Filter Bypass. The JDRP shall be equipped with the ability to bypass the filter separator.

2.14 Pumping accessories.

2.14.1 Hoses. The JDRP shall come equipped with 75 feet of fuel servicing hose and 100 feet of fuel suction hoses.

Fuel servicing hoses shall be hard/semi hard type, 2.5 inches in diameter and 75 feet long, intrinsically bonded. Hoses shall meet or exceed EI 1529 Grade II, Type C requirements. Servicing hose must be on an electric retractable hose reel with manual back up.

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Fuel receipt/suction hoses shall be hard/semi hard and provide sufficient NPSH to maintain performance requirements. Hoses shall be constructed in 10-foot-long sections. Fuel suction hoses shall meet or exceed MIL-PRF-370 type B or MIL-DTL-27516 specifications.

The JDRP shall be equipped with the following hose assemblies or equivalent:

a. 10ea - 4 inch, 150 PSI, 10 feet long Camlock fuel hose.

b. 1ea - 2.5 inch, 75 feet long, EI 1529 Grade II, Type C aircraft fueling hose coupled with 2.5 inch NPT ends on retractable hose reel.

2.14.2 Nozzle. The JDRP shall be equipped with an SAE AS5877 D3 nozzle to a MS24484 single point receptacle. The nozzle shall be equipped with a 60-mesh strainer. Cla-Val 341 GF-D3 or equivalent.

2.14.3 Adapters, fittings, and spare items. The JDRP shall be equipped with the following or equivalent items:

a. 1ea - 4-inch Female Camlock to 3-inch Male Camlock. GA40F30M or equivalent.

b. 2ea - 4-inch Female to 4-inch Female Camlock 90-degree elbow. DA40F40F90 or equivalent.

c. 2ea - 4-inch, Aluminum Female Cam and Groove Coupler x Male NPT Thread, type B. B400-AL-Buna or equivalent.

d. 2ea - 4-inch, Aluminum Male Cam and Groove Adapter x Male NPT Thread fitting, type F. F400-AL-Buna or equivalent.

2.14.4 Basket Strainer. The JDRP shall be equipped with a removable basket strainer for fuel receipt. The basket strainer shall be located on the suction line prior to the pump.

2.15 Manuals.

2.15.1 Technical manuals. The overall format for the manuals may be military specification, commercial specification, or a combination of both. Each technical manual shall have a title page. One complete set of manuals shall be packed with each JDRP. An additional two complete sets of manuals shall be sent to the procuring activity for stock.

The contractor shall submit digitized technical manuals and printed paper copies. The contractor may recommend the delivery of files developed in a "native" format as the final delivery medium. Native format includes MS Word, Framemaker, or Interleaf. However, all recommendations for native formatted data or graphics shall be approved by the Technical Order Manager prior to any work being performed.

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a. The contractor shall validate the technical manuals for accuracy prior to submission to the procuring activity for verification. The contractor shall submit one complete set to the procuring activity for verification 30 days before the first production inspection. Any changes or corrections noted by the procuring activity shall be corrected and updated pages or manuals shall be submitted to the procuring activity.

b. Once approved by the procuring activity and a Technical Order (TO) number has been assigned, the contractor shall pack two complete sets of both hard copy and CDs with each JDRP. All manuals provided must be JDRP specific and reflect the configuration of the JDRP as built. Generic manuals are not acceptable. One complete set of manuals in CD format shall be sent to the procuring activity for stock.

c. The contractor shall grant the United States Air Force a non-exclusive, non-assignable, royalty free U.S. Government (Government) Purpose License, to scan into CENTRA (the USAF repository) and to reproduce and distribute (either electronically or via hard medium) copies or facsimiles of manuals produced and distributed by the contractor for this CID. These rights extend to Government agencies only, and the data contained in the technical manuals are not to be sold, disclosed or otherwise provided to any other entity or entities outside the Government. The license shall remain in effect as long as the JDRP described in the required technical manuals remain under Government control and usage. In addition, the contractor shall grant permission for the Government to place an Air Force TO number and publication date as well as distribution, warning, handling, and destruction statements on the cover / title pages of the TO.

2.15.2 Operator's manual. The operator's manual shall include all information required for the safe and efficient operation of the JDRP, including the fire extinguishing equipment, and any special attachments or auxiliary equipment. The operator's manual shall include the following items at a minimum:

a. Location and function of all controls and instruments shall be illustrated and fully described.

b. Safety information that is consistent with the safety standards established by the Occupational Safety and Health Administration (OSHA).

c. Checks and adjustments in preparation for placing the JDRP in service upon receipt from the contractor.

d. Preparation for shipment or storage.

e. Warranty information and period of the warranty for the complete JDRP and for any component warranty that exceeds the warranty of the complete JDRP. Addresses and telephone numbers shall be provided for all warranty providers.

f. General description of and step-by-step instructions for the operation of the JDRP and its fuel pumping systems and auxiliary equipment.

g. Description of the post-operational procedures (such as draining or flushing).

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h. Checklists for the daily maintenance inspection and mission readiness checks that the operator is expected to perform, including basic troubleshooting procedures.

i. Schedules for required preventative maintenance and required periodic maintenance.

j. A line art drawing of the JDRP, including front, rear, left, and right side views, showing basic dimensions and weight.

2.15.3 Service manual. The service manual shall identify any special tools and test equipment required and shall cover troubleshooting and maintenance as well as minor and major repair procedures. The text shall contain performance specifications, tolerances, and fluid capacities; current, voltage, and resistance data; test procedures; and such illustrations and exploded views as may be required to permit proper maintenance by qualified mechanics. The manual shall contain an alphabetical subject index as well as a table of contents. The service manual shall contain at least the following, where applicable:

a. Fuel pumping system schematic.

b. Electrical schematic.

c. Schedules for required preventative maintenance and required periodic maintenance.

d. Location, procedure, and interval for parts of the JDRP and equipment that require lubrication.

2.15.4 Parts manual. The parts manual shall include illustrations and exploded views, as needed, to properly identify all parts, assemblies, subassemblies, and special equipment. All components of assemblies shown in illustrations or exploded views shall be identified by reference numbers that correspond to the reference numbers in the parts lists. All purchased parts shall be cross-referenced with the original manufacturer's name and part number. The parts identification manual shall provide the description and quantity of each item used per vehicle. The size, thread dimensions, torque specifications, and special characteristics shall be provided for all nonstandard nuts, bolts, screws, washers, grease fittings, and similar items. The manual shall contain a numerical index. The parts manual shall contain a list of all of the component vendor names, addresses, and telephone numbers referenced in the parts list.

2.16 Workmanship. The JDRP, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Workmanship objectives shall include freedom from blemishes, defects, burrs and sharp corners and edges; accuracy of dimensions, surface finish, and radii of fillets; thoroughness of welding, painting, and riveting; marking of parts and assemblies; and proper alignment of parts and tightness of assembly fasteners.

2.16.1 Bolted connections. Bolt holes shall be accurately punched or drilled and shall be deburred. Threaded fasteners shall be tight and shall not work loose during testing or service usage.

2.16.2 Riveted connections. Rivet holes shall be accurately punched or drilled and shall be deburred. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet

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heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the component.

2.16.3 Gear and lever assemblies. Gear and lever assemblies shall be properly aligned and meshed and shall be operable without interference, tight spots, loose spots, or other irregularities. Where required for accurate adjustment, gear assemblies shall be free of excessive backlash.

2.16.4 Cleaning. The JDRP shall be thoroughly cleaned. Loose, spattered, or excess solder; welding slag; stray bolts, nuts, and washers; rust; metal particles; pipe compound; and other foreign matter shall be removed during and after final assembly.

3. REGULATORY REQUIREMENTS.

3.1 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with 23.403 of the Federal Acquisition Regulation (FAR). However, used, rebuilt, or refurbished items shall not be provided.

3.2 Green Procurement Program. Green Procurement Program (GPP) is a mandatory federal acquisition program that focuses on the purchase and use of environmentally preferable products and services. GPP requirements apply to all acquisitions using appropriated funds, including services and new requirements. FAR 23.404(b) applies and states the GPP requires 100% of EPA designated product purchase that are included in the Comprehensive Procurement Guidelines list that contains recovered materials, unless the item cannot be acquired: a) competitively within a reasonable timeframe; b) meet appropriate performance standards, or c) at a reasonable price. The prime contractor is responsible for ensuring that all subcontractors comply with this requirement.

4. PRODUCT CONFORMANCE PROVISIONS.

The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace, modified as necessary to comply with the requirements herein. The Government reserves the right to require proof of such conformance.

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First production inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First production inspection. The first production JDRP shall be subjected to the analyses, demonstrations, examinations, and tests described in 4.5.1 through 4.5.13. The contractor shall

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provide or arrange for all test equipment and facilities. Except as otherwise specified, all testing in which the engine is operated shall be performed using JP-8 turbine fuel. The compatibility test shall be performed after successful completion of all other portions of the first production inspection.

4.3 Conformance inspection. Each production JDRP shall be subjected to the examination described in 4.5.1, 4.5.9, 4.5.10, 4.5.11, and 4.5.13.

4.4 Inspection requirements.

4.4.1 General inspection requirements. Apparatus used in conjunction with the inspections specified herein shall be laboratory precision type, calibrated at proper intervals to ensure laboratory accuracy.

4.4.2 Data. During all testing specified herein, at least the following data, unless not applicable, shall be recorded at intervals not to exceed 30 minutes. Additional data or shorter intervals shall be provided as appropriate for any specific test.

a. Date.

b. Time started.

c. Time finished.

d. Ambient temperature.

e. Ambient humidity.

4.4.3 Test rejection criteria. Throughout all tests specified herein, the JDRP shall be closely observed for the following conditions, which shall be cause for rejection.

a. Failure to conform to design or performance requirements specified herein or in the contractor's technical proposal.

b. Any spillage or leakage of any liquid, including fuel, lubricant, or hydraulic fluid, under any condition, except as allowed herein.

c. Thermal or structural failure of any component, including permanent deformation, or evidence of impending failure.

d. Evidence of excessive wear. If excessive wear is suspected, the original equipment manufacturer’s (OEM’s) specifications or tolerances shall be utilized for making a determination.

e. Evidence of corrosion or deterioration.

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f. Misalignment of components.

g. Conditions that present a safety hazard to personnel during operation, servicing, or maintenance.

h. Interference between the JDRP components or between the JDRP, the ground, and all required obstacles, with the exception of normal contact by the tires.

i. Evidence of undesirable mobility characteristics, including instability in handling during cornering, braking, and while traversing all required terrain.

j. Shutdown faults from:

(1) Engine cooling system.
(2) Engine lubrication system.
(3) Engine protective circuits.

4.5 Test methods.

4.5.1 Examination of product. Each JDRP shall be examined to verify compliance with the requirements herein prior to accomplishing any other demonstrations or tests listed in 4.5. A contractor-generated, Government-approved checklist (part of the test procedure) shall be used to identify each requirement not verified by an analysis, certification, demonstration, or test, and shall be used to document the examination results. Particular attention shall be given to materials, workmanship, dimensions, surface finishes, protective coatings and sealants and their application, welding, fastening, and markings. Proper operation of each JDRP function shall be verified. Certifications and analyses shall be provided in accordance with Table III. Each production JDRP shall be inspected to a Government-approved reduced version of the checklist.

TABLE III. Certifications and analyses.

Paragraph	Required Certifications and Analyses
2.3.3.2 Snow.	Contractor analysis of the snow load requirement (see 4.5.4.3.2).
2.3.4 Solar radiation.	Contractor certification that the JDRP performance is not adversely affected by full time exposure to solar radiation, such as those conditions encountered in desert environments.
2.3.5 Fungus.	Contractor certification that the materials used in construction of the JDRP are fungus resistant or suitably treated to resist fungus.

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Paragraph	Required Certifications and Analyses
2.5.1 Surface transportability.	Contractor surface transportability analysis (see 4.5.6.1.1) and certification that the JDRP is transportable via all modes of surface shipment (highway, rail, and water) in accordance with MIL-STD-1366, and shall be capable of withstanding the mechanical shock and vibration characteristics of highway, rail, and water transport.
2.5.2 Air transportability., 2.5.2.1 Shoring., and 2.5.2.3 Winch points.	Contractor air transportability analysis (see 4.5.6.2).
2.5.3 Tie downs.	Contractor tie down provision analysis (see 4.5.6.4.1).
2.5.4 Lifting provisions.	Contractor lifting provision analysis (see 4.5.6.5.1).
2.9 Engine and related equipment., 2.9.3 Engine lubrication system., 2.9.6.1 Engine primary fuels., 2.9.6.2 Engine alternate fuels., and 2.9.6.3 Engine emergency fuels.	Engine manufacturer certification that the engine is in accordance with all applicable requirements, including exhaust emissions standards and fuels. Engine manufacturer application approval for the engine and its installation, including lubrication system, and mounting system.

4.5.2 Noise test. The free field noise level shall not exceed 84 dBA at 40 feet from the exhaust on the aft JDRP centerline with the microphone 5 feet above the ground.

4.5.3 Electromagnetic interference test. A first production JDRP shall be tested in accordance with MIL-STD-461: RE102, and RS103 to demonstrate compliance with 2.2.5. Any engine fuel in accordance with 2.9.6.1 may be utilized.

4.5.4 Environmental testing.

4.5.4.1 High temperature storage and operation test. A first production JDRP shall be tested in accordance with MIL-STD-810, Method 501.6, Procedures I and II, to demonstrate compliance with the high temperature storage and operating requirements of 2.3.1 and 2.3.2. Test duration shall be one 24-hour cycle for each procedure beginning no less than two hours after test item temperature stabilization.

4.5.4.2 Low temperature storage and operation test. A first production JDRP shall be tested in accordance with MIL-STD-810, Method 502.6, Procedures I and II, to demonstrate compliance with the low temperature storage and operating requirements of 2.3.1 and 2.3.2, as well as the engine starting requirement of 2.9.1.5 and the alternator charging requirement of 2.10.1. Test duration shall be one 24-hour cycle for each procedure beginning no less than two hours after test item temperature stabilization.

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4.5.4.3 Precipitation.

4.5.4.3.1 Rain test. A first production JDRP shall be tested in accordance with MIL-STD-810, Method 506.6, Procedure I, to demonstrate compliance with 2.3.3.1.

4.5.4.3.2 Snow load analysis. An engineering analysis shall be performed to demonstrate compliance with the snow load requirement of 2.3.3.2, using a specific gravity of snow of 0.1 (Ref. 5.3 of MIL-STD-810, Part Three).

4.5.4.3.3 Ice accretion test. A first production JDRP shall be tested in accordance with MIL-STD-810, Method 521.4 with an ice thickness of 1.5-inches to demonstrate compliance with the ice accretion requirement of 2.3.3.3. The contractor shall identify those areas of the JDRP where ice removal is required prior to operation.

4.5.4.4 Salt fog test. A first production JDRP shall be tested in accordance with MIL-STD-810, Method 509.6, to demonstrate compliance with 2.3.6. Test duration shall be alternating 24-hour periods of salt fog exposure and drying conditions for 24-hour periods (two wet and two dry).

4.5.4.5 Sand and dust test. A first production JDRP shall be tested in accordance with MIL-STD-810, Method 510.6, Procedures I (12 hours) and II (90 minutes per side), to demonstrate compliance with 2.3.7.

4.5.5 Weight and dimension tests.

4.5.5.1 Weight and center of gravity test. The weight and center of gravity of a first production JDRP shall be measured to demonstrate compliance with the weight requirement of 2.4.

4.5.5.2 Dimension measurement. A first production JDRP shall be measured to demonstrate compliance with the dimensional requirements of 2.4.

4.5.6 Transportability verification.

4.5.6.1 Surface transportability verification.

4.5.6.1.1 Surface transportability analysis. An engineering analysis shall be performed to demonstrate compliance with 2.5.1. The engineering analysis shall utilize the data for road transportation in accordance with MIL-STD-810, Method 514.7, Table 514.7C-II.

4.5.6.2 Air transportability analysis. An engineering analysis shall be performed to demonstrate compliance with the air transportability requirements of 2.5.2, 2.5.2.1, 2.5.2.3, and 2.5.2.4 The analysis shall include the tie downs and all major components and their ability to withstand the accelerations specified in 2.5.2. The analysis shall demonstrate the accessibility and structural adequacy of the winch points specified in 2.5.2.3. The evaluation shall also include a dimensional analysis for the JDRP while traversing the ramp and while loaded aboard C-5, C-17, and C-130 aircraft.

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4.5.6.3 Equipment removal and reconfiguration demonstration. A first production JDRP shall be configured for transport on C-5, C-17, and C-130 aircraft and then reconfigured for operation to demonstrate compliance with 2.5.2.4. It shall be demonstrated that the forces required do not exceed those allowed in MIL-STD-1472.

4.5.6.4 Tie down provision verification.

4.5.6.4.1 Tie down provision analysis. An engineering analysis shall be performed to demonstrate compliance with the tie down provision requirements of 2.5.3.

4.5.6.4.2 Tie down provision test. A first production JDRP shall be tested to demonstrate compliance with the tie down provision requirements of 2.5.3.

4.5.6.5 Lifting provision verification.

4.5.6.5.1 Lifting provision analysis. An engineering analysis shall be performed to demonstrate compliance with the lifting provision requirements of 2.5.4.

4.5.6.5.2 Lifting provision test. A first production JDRP shall be tested to demonstrate compliance with the lifting provision requirements of 2.5.4.

4.5.7 Maintainability demonstration. All recommended preventive maintenance tasks shall be performed, and the task times shall be recorded. The recommended frequencies of the preventive maintenance tasks and the times recorded to accomplish the tasks shall be used to develop an expected value of preventive maintenance time per measure of use, such as calendar time or hours of operation. It shall be demonstrated that the forces required do not exceed those allowed in MIL-STD-1472. All preventive maintenance tasks recommended to be performed daily and at the routine PMI shall also be performed by personnel wearing arctic mittens.

4.5.8 Mobility tests. A first production JDRP shall be tested in accordance with Table VIII of SAE AS8090 to demonstrate compliance with the mobility requirements of 2.8. Note that the maximum towing force is 75 pounds per ton of gross weight, in accordance with 3.3.4 of SAE AS8090, rather than 50 pounds per ton of gross weight as stated in Table VIII of SAE AS8090.

4.5.9 Engine and pump endurance test. For first production inspection the JDRP engine and fuel transfer pump shall be subjected to seven continuous two-hour cycles consisting of one hour of operation followed by one hour of non-operation. For production conformance inspection the JDRP shall be subjected to one continuous hour of operation. The one hour of operation shall consist of the engine at idle with no flow for 15 minutes, followed by running the engine/pump to achieve 300 +0/-50 gpm through one discharge hose and nozzle for 45 minutes. Adequate NPSH shall be provided to the pump to support the flow rates. During each operational cycle, it shall be demonstrated that the nozzle pressure control system limits the pressure at the nozzle to no more than 55 psi regardless of flowrate. After one hour of continuous operation, the engine shall be turned off for one hour of non-operation. Upon completion of seven continuous two-hour cycles, an engine or pump failure or impending failure shall be cause for rejection. A failure is the event,

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or inoperable state, in which any item, or part of an item, does not, or would not, perform as previously specified. Any adjustment, repair, or replacement action, which results in degradation of system performance, shall be cause for rejection.

4.5.10 Nozzle pressure test. A test shall be conducted to verify the accuracy of the simulated nozzle pressure gauge and calibration of the venturi is within ± 5 psi of actual nozzle pressure.

4.5.11 High/low pressure pumping control test. A test shall be conducted to demonstrate that the flow and pressure control system is in compliance with 2.13.8. The test shall demonstrate that when in fuel servicing operations and utilizing one discharge hose, the primary nozzle pressure control system shall limit the nozzle pressure to 30 ± 5 psi when in low-pressure mode and 50 ± 5 psi when in high-pressure mode. It shall be demonstrated that the secondary pressure control system for each discharge hose shall limit the nozzle pressure to 55 psi when operating in either pressure mode.

4.5.12 Service life analysis. An engineering analysis shall be performed to demonstrate compliance with the service life requirement of 2.6.5.

4.5.13 Hydrostatic testing. Each JDRP shall be hydrostatically tested to demonstrate compliance with the hydrostatic system pressure requirement in 2.13.4.

5. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES.

6.1 Source of documents.

6.1.1 Department of Defense and Federal documents. Department of Defense and Federal documents, except for GOST 10227-86, are available online at https://quicksearch.dla.mil/. A copy of GOST 10227-86 can be obtained from the Procuring Contracting Officer (PCO).

6.1.2 Air Force Instructions (AFI). Copies of AFIs are available online at https://www.e-publishing.af.mil/.

6.1.3 Code of Federal Regulations (CFR). Copies of the CFR are available online at https://www.ecfr.gov/.

6.1.4 Energy Institute (EI) documents. Copies of EI documents may be obtained online at https://www.energyinst.org/.

6.1.5 Environmental Protection Agency (EPA). Copies of EPA documents may be obtained online at https://www.epa.gov.

6.1.6 Federal Acquisition Regulations (FAR). Copies of FARs may be obtained online at https://www.acquisition.gov/?q=browsefar.

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6.1.7 American Society for Testing and Materials (ASTM) documents. Copies of ASTM standards may be obtained online at https://www.astm.org/.

6.1.8 American Welding Society (AWS) documents. Copies of AWS standards may be obtained online at https://www.aws.org/.

6.1.9 International Organization for Standardization (ISO) documents. Copies of ISO documents may be obtained online at https://www.iso.org/home.html.

6.1.10 Occupational Safety and Health Association (OSHA) standards. Copies of OSHA standards may be obtained online at https://www.osha.gov/.

6.1.11 North Atlantic Treaty Organization (NATO) documents. Copies of NATO documents may be obtained online at https://www.nato.int/.

6.1.12 National Electrical Manufacturers Association (NEMA) documents. Copies of NEMA documents may be obtained online at https://www.nema.org.

6.1.13 SAE International documents. Copies of SAE standards may be obtained online at https://www.sae.org/.

6.1.14 Society for Protective Coatings (SSPC) documents. Copies of SSPC documents may be obtained online at https://sspc.org/.

6.1.15 Technology and Maintenance Council (TMC) documents. Copies of TMC standards may be obtained online at https://www.trucking.org/Technology_Council.aspx.

6.1.16 Tire and Rim Association (TRA) documents. Copies of TRA standards may be obtained online at https://www.us-tra.org/.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this CID.

6.3 Definitions.

6.3.1. Alternate fuel. A fuel that can be used in place of the primary fuel with a possible loss of efficiency, engine adjustments may be necessary. The use of an alternate fuel may result in a change of maintenance or overhaul cost.

6.3.2 Belgian-block. A Belgian-block course consists of a very rough, undulating, hard surface unaffected by changes in climatic conditions. The standard is found at Aberdeen Proving Grounds, Maryland.

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6.3.3 Cobblestone. A naturally rounded stone larger than a pebble and smaller than a boulder; *especially*: such a stone used in paving a street or in construction.

6.3.4 Commercial item.

(1) Any item, other than real property, that is of a type customarily used by the general public or by non-Governmental entities for purposes other than Governmental purposes, and—

(i) Has been sold, leased, or licensed to the general public; or

(ii) Has been offered for sale, lease, or license to the general public;

(2) Any item that evolved from an item described in paragraph (1) of this definition through advances in technology or performance and that is not yet available in the commercial marketplace, but will be available in the commercial marketplace in time to satisfy the delivery requirements under a Government solicitation;

(3) Any item that would satisfy a criterion expressed in paragraphs (1) or (2) of this definition, but for—

(i) Modifications of a type customarily available in the commercial marketplace; or

(ii) Minor modifications of a type not customarily available in the commercial marketplace made to meet Federal Government requirements. Minor modifications means modifications that do not significantly alter the non-Governmental function or essential physical characteristics of an item or component, or change the purpose of a process. Factors to be considered in determining whether a modification is minor include the value and size of the modification and the comparative value and size of the final product. Dollar values and percentages may be used as guideposts, but are not conclusive evidence that a modification is minor;

(4) Any combination of items meeting the requirements of paragraphs (1), (2), (3), or (5) of this definition that are of a type customarily combined and sold in combination to the general public;

(5) Installation services, maintenance services, repair services, training services, and other services if—

(i) Such services are procured for support of an item referred to in paragraph (1), (2), (3), or (4) of this definition, regardless of whether such services are provided by the same source or at the same time as the item; and

(ii) The source of such services provides similar services contemporaneously to the general public under terms and conditions similar to those offered to the Federal Government;

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(6) Services of a type offered and sold competitively in substantial quantities in the commercial marketplace based on established catalog or market prices for specific tasks performed or specific outcomes to be achieved and under standard commercial terms and conditions. For purposes of these services—

(i) "Catalog price" means a price included in a catalog, price list, schedule, or other form that is regularly maintained by the manufacturer or vendor, is either published or otherwise available for inspection by customers, and states prices at which sales are currently, or were last, made to a significant number of buyers constituting the general public; and

(ii) "Market prices" means current prices that are established in the course of ordinary trade between buyers and sellers free to bargain and that can be substantiated through competition or from sources independent of the offerors.

(7) Any item, combination of items, or service referred to in paragraphs (1) through (6) of this definition, notwithstanding the fact that the item, combination of items, or service is transferred between or among separate divisions, subsidiaries, or affiliates of a contractor; or

(8) A nondevelopmental item, if the procuring agency determines the item was developed exclusively at private expense and sold in substantial quantities, on a competitive basis, to multiple State and local Government. (Reference the Federal Acquisition Regulation (FAR) 2.101)

6.3.5 Common hand tool. A non-powered tool that is likely to be found in a typical mechanic's toolbox. Common hand tools include open end, boxed end, combination, socket (both 6- and 12-point in both standard and deep-well), and hex key wrenches, in SAE sizes up to and including 1-inch and metric sizes up to and including 25-mm; ratchet handles, extensions, and swivels; slotted and Phillips-head screwdrivers; regular and snap-ring pliers; and a ball-peen hammer.

6.3.6 Cross-country terrain. Cross-country terrain consists of unimproved open fields, broken ground, loose sand, and gravel road encountered by tactical vehicles supporting an army in the field.

6.3.7 Emergency fuel. A fuel that can be used when the primary and alternate fuels are not available. This fuel is not for use on a continuing basis, but is to be only when the primary and alternate fuel and equipment fuels are unobtainable and operation is mission essential. Engine life could be severely impacted.

6.3.8 Gravel road. A gravel road is a level or rolling gravel trail.

6.3.9 Primary fuel. The fuel or fuels used during tests to demonstrate system performance (contract compliance). Substitutions are presented as Alternate and Emergency fuels.

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6.3.10 Special tool. A tool that is not commercially and readily available from a source other than the JDRP contractor.

6.4 Key words.

Aircraft fuel servicing
Aircraft refueling
Air transportable
Mobile aircraft fuel pump
Trailer mounted

Custodian:
Air Force – 184

Preparing Activity:
Air Force – 184

Agent:
Air Force – 110

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